Why 13?

The Fano plane, the split Cayley hexagon GH$(2,2)$,
and the Hecke algebra $H(G_2(2), \mathrm{PSL}(3,2))$

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Abstract

The Hecke algebra $H(G_2(2), \mathrm{PSL}(3,2))$ — discovered in the "Meinhold-72" sessions — has a two-dimensional irreducible representation generated by two matrices A, B with

$$A^2 = I, \qquad B^2 = 13\, I, \qquad AB + BA = -4I.$$

Where does the 13 come from? This paper explains *everything* needed to answer that: what a Fano plane is (six definitions), what the split Cayley hexagon is (four definitions), how they are related (Fano planes are generalized triangles, hexagons are generalized hexagons; PSL$(3,2)$ sits inside $G_2(2)$ in exactly 72 ways, each fixing a chiral Fano structure), and finally a complete combinatorial derivation of

$$13 \;=\; 7 + 2 \cdot 3,$$

where 7 is an orbital valency and $3 = \sqrt{k-\mu}$ is an eigenvalue of the strongly regular graph SRG$(36, 21, 12, 12)$ on the 36 Fano structures of one chirality. All computational claims are verified in GAP (`why13_fano.g`, `why13_orbitals.g`).

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1 Introduction: the question

The group $G = G_2(2)$ of order $12096 = 2^6 \cdot 3^3 \cdot 7$ acts primitively on 63 points (`PrimitiveGroup(63,2)` in GAP). It contains a subgroup $K \cong \mathrm{PSL}(3,2)$ of order 168 and index 72, unique up to conjugacy. The double cosets $K\backslash G/K$ give a Hecke algebra

$$H(G,K) \;\cong\; \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}$$

of rank 8, with canonical basis $T_1, \ldots, T_8$ of valencies $1, 1, 7, 7, 7, 7, 21, 21$. In the 2-dimensional Wedderburn block, the images $A = \rho(T_2)$ and $B = \rho(T_3)$ satisfy

$$A^2 = I, \qquad B^2 = 13\, I, \qquad AB + BA = -4I.$$

The relation $B^2 = 13I$ is forced by the structure constants: in the algebra,

$$T_3 \cdot T_3 \;=\; 7\, T_1 + 2\, T_8, \tag{1}$$

and T_8 is *central* with $\rho(T_8) = 3I$, so $B^2 = 7I + 2 \cdot 3I = 13I$. The question "why 13?" therefore splits into:

1. Why does T_3^2 hit only the identity and the central element T_8, with coefficients 7 and 2? (Combinatorics of the orbital graphs.)

2. Why does the central element act by the scalar 3? (Spectral theory of a strongly regular graph.)

Both questions are about the geometry of *Fano planes living inside the split Cayley hexagon*. So we start from scratch.

2 Incidence geometry and the Fano plane

2.1 Incidence structures and projective planes

Definition 2.1 (Incidence structure)**.** An *incidence structure* is a triple $(\mathcal{P}, \mathcal{L}, \mathrm{I})$ where $\mathcal{P}$ is a set of *points*, $\mathcal{L}$ a set of *lines*, and $\mathrm{I} \subseteq \mathcal{P} \times \mathcal{L}$ an *incidence relation*. We write $p \in \ell$ for $(p, \ell) \in \mathrm{I}$ when lines are identified with their point sets. A *flag* is an incident pair (p, ℓ); an *antiflag* is a non-incident pair.

Definition 2.2 (Projective plane)**.** A *projective plane* is an incidence structure such that:

(P1) any two distinct points lie on exactly one common line;

(P2) any two distinct lines meet in exactly one common point;

(P3) (non-degeneracy) there exist four points, no three collinear.

A plane has *order* n if some (equivalently every) line has $n + 1$ points; then every point is on $n + 1$ lines, and there are $n^2 + n + 1$ points and $n^2 + n + 1$ lines.

Example 2.3. The real projective plane $\mathbb{RP}^2$ satisfies (P1)–(P3) but is infinite. The smallest possible order is $n = 2$: lines of 3 points, 7 points, 7 lines.

Non-example 2.4. The *near-pencil*: points $\{1, 2, 3, 4\}$, lines $\{1, 2, 3\}, \{1, 4\}, \{2, 4\}, \{3, 4\}$. It satisfies (P1) and (P2) but fails (P3): every set of four points contains three on the line $\{1, 2, 3\}$ or includes point 4 twice. Degenerate "planes" like this are exactly what (P3) excludes.

2.2 The Fano plane: six definitions

Definition 2.5 (Fano plane, axiomatic)**.** The *Fano plane* is the (unique, by Theorem 2.13) projective plane of order 2: 7 points, 7 lines, 3 points per line, 3 lines per point.

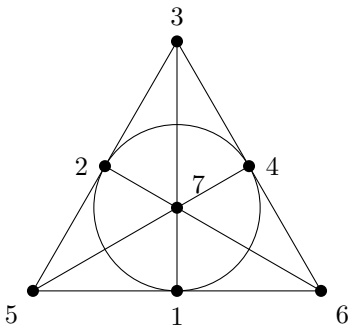


Figure 1: The Fano plane with the difference-set labelling $\ell_i = \{i + 1, i + 2, i + 4\}$ (mod 7). The seven lines are the three sides $\{5, 6, 1\}, \{2, 3, 5\}, \{3, 4, 6\}$, the three medians $\{7, 1, 3\}, \{4, 5, 7\}, \{6, 7, 2\}$, and the circle $\{1, 2, 4\}$. Check: each is a translate of $\{1, 2, 4\}$ mod 7.

Definition 2.6 (Fano plane as $\mathrm{PG}(2, 2)$)**.** Let $V = \mathbb{F}_2^3$. Points are the 7 one-dimensional subspaces (= nonzero vectors, since $|\mathbb{F}_2^\times| = 1$); lines are the 7 two-dimensional subspaces (= kernels of nonzero linear functionals). A point lies on a line iff the vector lies in the subspace. Equivalently: three nonzero vectors u, v, w are collinear iff $u + v + w = 0$.

Example 2.7. Write each point $i \in \{1, \dots, 7\}$ in binary: $1 = 001$, $2 = 010$, $3 = 011$, $4 = 100$, $5 = 101$, $6 = 110$, $7 = 111$. Then $\{1, 2, 3\}$ is a line since $001 \oplus 010 \oplus 011 = 000$. (This binary labelling is the Hamming-code labelling; it differs from the difference-set labelling of Figure 1 by a relabelling.)

Definition 2.8 (Fano plane as a Steiner system)**.** A *Steiner system* $S(2, 3, 7)$ is a collection of 3-element blocks of a 7-set such that every 2-subset lies in exactly one block. Any $S(2, 3, 7)$ is a Fano plane: axiom (P1) is the Steiner condition, and (P2) follows by counting — two blocks meeting in ≥ 2 points would violate uniqueness, and two disjoint blocks would leave the pairs across them uncovered by a count: each block covers 3 pairs, 7 blocks cover $21 = \binom{7}{2}$ pairs exactly once, and through a point p outside a block ℓ the three lines pq ($q \in \ell$) are distinct, so they exhaust the 3 lines through p; hence no line through p misses ℓ.

Definition 2.9 (Fano plane from a perfect difference set)**.** $D = \{1, 2, 4\} \subset \mathbb{Z}/7$ is a *perfect difference set*: every nonzero residue mod 7 arises exactly once as a difference $d - d'$ with $d, d' \in D$. (Check: 2−1=1, 4−2=2, 4−1=3, 1−4=4, 2−4=5, 1−2=6.) The translates $\ell_i = i + D$ ($i \in \mathbb{Z}/7$) are the lines of a Fano plane on the points $\mathbb{Z}/7$: the pair $\{x, y\}$ lies on ℓ_i iff $x - i, y - i \in D$, and since $x - y$ has a *unique* representation as a difference of elements of D, there is exactly one such i.

Non-example 2.10. $D' = \{0, 1, 2\}$ is *not* a perfect difference set: its differences are $\pm 1, \pm 2$ only, so 3 and 4 never occur. Concretely, the translates of D' give 7 triples in which the pair $\{0, 3\}$ lies on *no* line — axiom (P1) fails. "Seven triples covering each point three times" is not enough; the difference structure is essential.

Definition 2.11 (Fano plane as octonion multiplication mnemonic)**.** Label the seven imaginary units of the octonions $e_1, \dots, e_7$ and decree $e_i e_{i+1} = e_{i+3}$ (indices mod 7), extended by the rules $e_i^2 = -1$ and anticommutativity. The triples $\{i, i{+}1, i{+}3\}$ are the lines of a Fano plane (the difference set $\{0, 1, 3\}$ works just like $\{1, 2, 4\}$: its differences $1, 3, 2, 6, 4, 5$ cover everything once). Each oriented line is a copy of the quaternions. The Fano plane with oriented lines *is* the multiplication table of the octonions.

Definition 2.12 (Fano plane via the Heawood graph)**.** The *incidence graph* (Levi graph) of an incidence structure is the bipartite graph on $\mathcal{P} \sqcup \mathcal{L}$ with edges the flags. The Fano plane is the incidence structure whose incidence graph is the *Heawood graph*: the unique bipartite cubic graph of girth 6 on 14 vertices (Figure 2). In the language of Section 3: the Fano plane is the unique *generalized triangle of order* $(2, 2)$. This is the definition that puts it in the same family as the split Cayley hexagon.

Theorem 2.13 (Uniqueness)**.** *All projective planes of order* 2 *are isomorphic.*

Proof. Let Π be a plane of order 2; fix a line $\ell = \{a, b, c\}$ and a point $d \notin \ell$ (exists by (P3)). The lines da, db, dc are distinct and account for all 3 lines through d; each has one more point, giving points $e \in da$, $f \in db$, $g \in dc$, and these are distinct (two of them equal would put two lines through the pair $\{d, \cdot\}$). So $\Pi = \{a, \dots, g\}$, 7 points, and the incidences listed so far are forced. The remaining 3 lines each must join e, f, g to the points of ℓ not already joined to them; one checks there is exactly one way to complete: e.g. the line through e and b cannot contain a (that's ℓ? no — $a, b \in \ell$, and $e \notin \ell$, so the line $eb \neq \ell$ and meets ℓ only in b), cannot contain d or f (lines da, db already pass through those pairs), so its third point is forced to be g or c; and ec

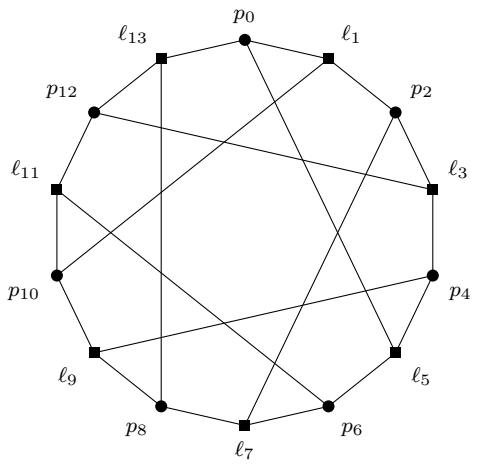


Figure 2: The Heawood graph: a 14-cycle plus the chords $\{0,5\},\{2,7\},\{4,9\},\{6,11\},\{8,13\},\{10,1\},\{12,3\}$. Circles are points, squares are lines of the Fano plane; girth 6 encodes "no digons": two points on two common lines would give a 4-cycle.

would leave the pair $\{e,g\}$ uncoverable. Following these forced choices yields a unique incidence table, so any two order-2 planes admit an isomorphism mapping the chosen configurations to each other. □

Fact 2.14 (Automorphisms)**.** $\mathrm{Aut}(\text{Fano}) \cong \mathrm{GL}(3,2) = \mathrm{PSL}(3,2) \cong \mathrm{PSL}(2,7)$, the simple group of order $168 = 7{\cdot}6{\cdot}4$. From Definition 2.6, any linear automorphism of $\mathbb{F}_2^3$ permutes the points and lines; the order is $(2^3{-}1)(2^3{-}2)(2^3{-}2^2) = 7{\cdot}6{\cdot}4$. Conversely every plane automorphism is induced by a linear map because a plane automorphism is determined by its action on a quadrilateral (the proof of Theorem 2.13 shows a quadrilateral determines everything) and $\mathrm{GL}(3,2)$ is transitive on quadrilaterals. The action on flags (21 of them) is transitive, as is the action on antiflags ($42 = 7 \cdot 6$ of them). *Remember these numbers:* 21 *and* 42*.*

3 Generalized polygons

3.1 Definition and first examples

Definition 3.1 (Generalized n-gon)**.** A *generalized n-gon* is an incidence structure whose incidence graph Γ (bipartite, on points $\sqcup$ lines) satisfies:

(G1) $\mathrm{diam}(\Gamma) = n$;

(G2) $\mathrm{girth}(\Gamma) = 2n$;

(G3) (thickness) every vertex has degree ≥ 3.

It has *order* (s,t) if every line has $s+1$ points and every point is on $t+1$ lines. (Regularity is automatic from (G1)–(G3) for $n \geq 3$; we take it as given.)

The girth condition says: no "short circuits." A cycle of length $2k < 2n$ in the incidence graph is a closed configuration of k points and k lines — an ordinary k-gon. So a generalized

n-gon contains no ordinary k-gons for $k < n$, while diameter n plus girth $2n$ forces every pair of elements to lie in a common ordinary n-gon (an *apartment*).

Proposition 3.2. *Generalized 3-gons of order (s, s) are exactly the projective planes of order s.*

Proof. ($\Leftarrow$) In a projective plane's incidence graph, a point and a line are at distance 1 (incident) or 3 (join p to any $q \in \ell$, then to ℓ); two points are at distance 2 (via their unique joining line); likewise two lines. So diameter is 3. A 4-cycle would be two points on two common lines, contradicting (P1); so girth is $6 = 2 \cdot 3$ (hexagons exist: triangles of the plane). (P3) gives thickness.

($\Rightarrow$) Girth 6 means no 4-cycles: two points have at most one common line. Diameter 3 means a point p and non-incident line ℓ are at distance ≤ 3, and bipartiteness makes that distance odd, hence exactly 3: there is a line through p meeting ℓ. Applying this to "two points p, q": they are at even distance ≤ 3, hence 2: a common line exists. Combining: exactly one common line, which is (P1); dually (P2); thickness gives (P3). □

Fact 3.3 (Feit–Higman, 1964)**.** Finite thick generalized n-gons exist only for $n \in \{2, 3, 4, 6, 8\}$.

So after projective planes (n=3) and generalized quadrangles (n=4), the next family is the generalized *hexagons* (n=6). The Fano plane is the smallest thick generalized 3-gon; the split Cayley hexagon $\mathrm{GH}(2, 2)$ is the smallest thick generalized 6-gon. They are siblings in the same Tits family.

3.2 Counting in a generalized hexagon

Lemma 3.4 (Point count)**.** *A generalized hexagon of order (s, t) has $(1+s)(1+st+s^2t^2)$ points and $(1+t)(1+st+s^2t^2)$ lines. For $(s, t) = (2, 2)$: 63 points and 63 lines.*

Proof. Fix a line ℓ_0 and BFS in the incidence graph. ℓ_0 has $1 + s$ points. Each such point is on t further lines; no two of these lines share a point other than on ℓ_0 (a digon or quadrilateral would result, violating girth 12), so at distance 2 from ℓ_0 we get $(1 + s)t$ lines, each carrying s new points: $(1 + s)ts$ points at distance 3 (in the point-counting sense, distance 2 in the collinearity graph from ℓ_0's points). Repeat once more: $(1 + s)ts \cdot t$ lines, then $(1 + s)tsts$ points; girth 12 keeps everything distinct, and diameter 6 stops the growth there. Total points:

$$(1 + s)\,(1 + st + s^2t^2).$$

For $s = t = 2$: $3 \cdot (1 + 4 + 16) = 3 \cdot 21 = 63$. □

3.3 The collinearity graph and its spectrum

Definition 3.5 (Collinearity graph)**.** The *collinearity graph* of an incidence structure has the points as vertices, two joined iff they lie on a common line. For a generalized hexagon of order $(2, 2)$ it is regular of valency $s(t + 1) = 6$ (each point: 3 lines, 2 other points on each).

Proposition 3.6. *The collinearity graph of a generalized hexagon of order $(2, 2)$ is distance-regular with intersection array $\{6, 4, 4;\ 1, 1, 3\}$, diameter 3, and distance distribution $1 + 6 + 24 + 32 = 63$.*

Proof. Fix p. Distance 1: the 6 points collinear with p. From $q \perp p$: one of q's three lines is the line pq (containing 1 neighbour of p besides itself, namely the third point); the girth-12 condition prevents any other line through q from meeting the lines through p again. Count: $c_1 = 1$ (each distance-1 point sees p via 1 edge... precisely: $a_1 = 1$, the third point of the line pq), $b_1 = 4$ (the other two lines through q carry 2 new points each). So $|\Gamma_2(p)| = 6 \cdot 4/c_2$ with $c_2 = 1$: a point at distance 2 from p is joined to exactly one neighbour of p, because two such joins would create a quadrilateral (girth $12 > 8$). Hence 24 points at distance 2, each with $a_2 = 1$, $b_2 = 4$, and $|\Gamma_3(p)| = 24 \cdot 4/c_3$. The remaining count $63 - 1 - 6 - 24 = 32$ forces $c_3 = 3$, i.e. each point at distance 3 is joined to exactly 3 points at distance 2 — geometrically, p and a point opposite to it are joined by 3 paths, one through each line of p. □

Proposition 3.7 (Spectrum)**.** *The collinearity graph of* $\mathrm{GH}(2,2)$ *has spectrum*

$$6^1, \quad 3^{21}, \quad (-1)^{27}, \quad (-3)^{14}.$$

Proof. The eigenvalues of a distance-regular graph are those of the tridiagonal intersection matrix

$$L = \begin{pmatrix} 0 & 6 & 0 & 0 \\ 1 & 1 & 4 & 0 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 3 & 3 \end{pmatrix}, \qquad \det(\lambda I - L) = \lambda^4 - 5\lambda^3 - 15\lambda^2 + 45\lambda + 54 = (\lambda-6)(\lambda-3)(\lambda+1)(\lambda+3).$$

Multiplicities m_θ solve $\sum m = 63$, $\sum m\theta = 0$ (trace), $\sum m\theta^2 = 63 \cdot 6$ (closed 2-walks), $\sum m\theta^3 = 63{\cdot}6{\cdot}a_1 = 378$ (closed 3-walks, $a_1 = 1$ triangle-mate per edge end). Solving: $(m_6, m_3, m_{-1}, m_{-3}) = (1, 21, 27, 14)$. Check: $1 + 21 + 27 + 14 = 63$; $6 + 63 - 27 - 42 = 0$; $36 + 189 + 27 + 126 = 378$; $216 + 567 - 27 - 378 = 378$. *Remember* $m_{-3} = 14$*: it is the dimension of the* $G_2(2)$*-irreducible* χ_9 *that will carry the entire "why 13" story.* □

Remark 3.8. The permutation character of $G_2(2)$ on 63 points decomposes as $1+\chi_{21}+\chi_{27}+\chi_{14}$ (multiplicity-free, which is why the 63-point scheme is commutative); the eigenspace dimensions in Proposition 3.7 are exactly these degrees.

4 The split Cayley hexagon $\mathrm{GH}(2,2)$: four definitions

4.1 Definition A: Tits' construction on the parabolic quadric

Definition 4.1 (Split Cayley hexagon, quadric model)**.** Let $Q(6,q)$ be the parabolic quadric in $\mathrm{PG}(6,q)$, say $x_0x_4 + x_1x_5 + x_2x_6 = x_3^2$. The *split Cayley hexagon* $\mathrm{H}(q)$ has:

- **points**: all $q^5 + q^4 + \cdots + 1$ points of $Q(6,q)$ — for $q = 2$, all 63 of them;
- **lines**: a special class of lines of $Q(6,q)$, namely those whose Grassmann coordinates satisfy six explicit linear equations (Tits 1959). Not all lines on the quadric qualify: $Q(6,2)$ carries 315 lines, of which exactly 63 are hexagon lines.

4.2 Definition B: split octonions

Definition 4.2 (Split octonion model)**.** Let $\mathbb{O}'$ be the *split octonion algebra* over $\mathbb{F}_q$ (the unique octonion algebra over a finite field; it has zero divisors). Points of $\mathrm{H}(q)$ are the 1-spaces $\langle x \rangle$ of trace-zero elements with $x^2 = 0$; lines are the 2-spaces W of such elements with $W \cdot W = 0$ (all

products vanish). Incidence is containment. The automorphism group of $\mathbb{O}'$ is the Chevalley group $G_2(q)$ — this definition makes $G_2(q) \leq \mathrm{Aut}(\mathrm{H}(q))$ manifest, exactly as Definition 2.11 makes the (compact) octonions visible in the Fano plane.

Remark 4.3 (The Fano–octonion echo)**.** The Fano plane is the multiplication chart of the *compact* octonions (Definition 2.11); the split Cayley hexagon is the null geometry of the *split* octonions. Both incidence geometries are shadows of the same exceptional algebra, which is the deepest a-priori reason to expect Fano planes inside the hexagon.

4.3 Definition C: coset geometry / the model we compute with

Definition 4.4 (Coset geometry model)**.** Let $G = G_2(2)$, of order 12096, acting on the 63 cosets of a maximal parabolic P_1 (this is `PrimitiveGroup(63,2)`). Points of the hexagon are the 63 cosets. The point stabilizer has orbits of lengths $1, 6, 24, 32$ on points (matching Proposition 3.6); the valency-6 orbital is the collinearity graph. *Lines are recovered combinatorially as the triangles of the collinearity graph* (Proposition 4.5).

Proposition 4.5. *In the collinearity graph of* $\mathrm{GH}(2,2)$*, every edge lies in exactly one triangle, and the* 63 *triangles are exactly the lines.*

Proof. Each line $\{x, y, z\}$ is a triangle. Conversely, a triangle not contained in a line would be three points pairwise collinear via three distinct lines — an ordinary triangle in the hexagon, i.e. a 6-cycle in the incidence graph, contradicting girth 12. For "exactly one": an edge xy lies on the unique line through x, y (collinear points have a unique joining line, else a digon, i.e. 4-cycle, appears), whose third point completes the unique triangle. *Verified in GAP* (`why13_fano.g`): 189 edges, each with exactly 1 common neighbour; 63 triangles; 3 lines per point; collinearity-graph diameter 3. □

4.4 Definition D: symplectic model in even characteristic

Definition 4.6 ($W(5,q)$ model, q even)**.** For q even, the quadric $Q(6,q)$ has a nucleus, and projecting from it identifies $Q(6,q)$ with $\mathrm{PG}(5,q)$ equipped with a symplectic form: the hexagon $\mathrm{H}(q)$ embeds in the symplectic space $W(5,q)$. So for $q = 2$ the 63 points of $\mathrm{GH}(2,2)$ *are* the 63 nonzero vectors of $\mathbb{F}_2^6$, and the hexagon lines are 63 of the totally isotropic lines. This embedding is special to even characteristic.

Fact 4.7 (Cohen–Tits)**.** Up to isomorphism there are exactly two generalized hexagons of order $(2,2)$: the split Cayley hexagon $\mathrm{H}(2)$ and its point–line dual $\mathrm{H}(2)^D$, and they are *not* isomorphic (although each is, of course, a generalized hexagon with the same parameters). Both have automorphism group $G_2(2)$. *We work throughout with the one whose collinearity graph is the valency-*6 *orbital of* `PrimitiveGroup(63,2)`.

5 Hecke algebras of orbitals

Definition 5.1 (Orbitals and the centralizer algebra)**.** Let G act transitively on a finite set Ω, $|\Omega| = n$. The *orbitals* are the G-orbits $D_1, \ldots, D_r$ on $\Omega \times \Omega$, with $D_1 = \{(\omega, \omega)\}$ the diagonal. The number r is the *rank*. Each orbital has a 0/1 adjacency matrix A_i; the span $\langle A_1, \ldots, A_r \rangle$ is closed under multiplication:

$$A_i A_j = \sum_k c_{ij}^k A_k, \qquad c_{ij}^k = \#\{z : (x,z) \in D_i,\ (z,y) \in D_j\} \ \text{ for any fixed } (x,y) \in D_k.$$

This is the *centralizer algebra* (or Hecke algebra $H(G, G_\omega)$ — for $\Omega = G/K$ it is the double-coset algebra $\mathbb{C}[K\backslash G/K]$). The *valency* k_i of D_i is the out-degree of its orbital digraph; the *suborbits* are the G_ω-orbits on Ω, in bijection with orbitals. The *paired* orbital is $D_i^\top = \{(y, x) : (x, y) \in D_i\}$; D_i is *symmetric* if $D_i^\top = D_i$.

Example 5.2. $G_2(2)$ on 63 points: rank 4, suborbits $1, 6, 24, 32$, all symmetric, so the centralizer algebra is commutative — this is the association scheme of Proposition 3.6.

Lemma 5.3 (Symmetry criterion)**.** *D_i is symmetric iff $c_{ii}^1 = k_i$ (the coefficient of the identity class in $T_i \cdot T_i$ equals the valency); if D_i is not symmetric then $c_{ii}^1 = 0$ and the identity coefficient k_i appears in $T_i \cdot T_{i^\top}$ instead.*

Proof. c_{ii}^1 counts z with $(x, z) \in D_i$ and $(z, x) \in D_i$ for fixed x. If D_i is symmetric this is all k_i out-neighbours of x; if not, then $(x, z) \in D_i \Rightarrow (z, x) \in D_{i^\top} \neq D_i$, and orbitals are disjoint, so the count is 0. □

Remark 5.4. This is how one reads symmetry off a multiplication table. In the Meinhold-72 table, $T_3T_3 = 7T_1 + 2T_8$ and $T_5T_5 = 7T_1 + 2T_8$ contain T_1 with full valency 7 — symmetric; while $T_4T_4 = 4T_6 + T_8$ has no T_1 at all, and $T_4T_6 = 7T_1 + 2T_8$ — so $T_4^\top = T_6$, a genuinely *directed* pair. (This corrects an annotation in `MEINHOLD-HECKE.md` which listed T_3, T_6 as the asymmetric pair.)

6 The 72 Fano structures inside the hexagon

Everything in this section is computed and verified in `why13_fano.g` / `why13_orbitals.g`.

6.1 $\mathrm{PSL}(3, 2)$ inside $G_2(2)$

Proposition 6.1. *$G = G_2(2)$ has a unique conjugacy class of subgroups of order 168; the representative $K \cong \mathrm{PSL}(3, 2)$, of index 72. Its normalizer $N_G(K)$ has order 336, so the class has $12096/336 = 36$ unordered members but K has 72 cosets. The class does* not *split inside the simple subgroup $G' = \mathrm{PSU}(3, 3)$ of index 2: GAP confirms (matching the ATLAS, which lists $\mathrm{L}_2(7)$ as maximal in $\mathrm{U}_3(3)$ in a single class) that $N_{G'}(K) = K$, the normalizing involution lies* outside *G', and the G'-class still has $6048/168 = 36$ members. What* does *split is the coset space: G' has two orbits of 36 on the 72 cosets of K (GAP-verified), swapped by the outer involution of $G_2(2) = \mathrm{PSU}(3, 3){:}2$. Each of the 36 subgroups contributes exactly one coset to each orbit — its two orientations (Proposition 6.7).*

6.2 What K stabilizes: a chiral Fano structure

Proposition 6.2 (Orbits of K)**.** *K has point-orbits of sizes $21 + 42$ on the hexagon's 63 points and line-orbits of sizes $21 + 7 + 7 + 28$ on the 63 lines. Write P for the 21-point orbit, S_A and S_B for the two 7-line orbits. Then:*

1. *every line of S_A and of S_B is fully contained in P (intersection size 3), and each of S_A, S_B* partitions *P into 7 triples — two spreads of the same 21 points;*
2. *lines in the 21-line orbit meet P in exactly 1 point; lines in the 28-orbit miss P entirely;*
3. *the pairwise collinearity-graph distances within P are 1 (42 pairs), 2 (84), 3 (84): P is a 21-point, $189/\ldots$ — precisely, P induces a graph of valency 4 whose edges are the 42 pairs at distance 1.*

Proposition 6.3 (The Fano structure on the spreads)**.** *Define an incidence structure* Φ*:* points = *the* 7 *lines of* S_A*,* lines = *the* 7 *lines of* S_B*, with* $a \in S_A$ *incident to* $b \in S_B$ *iff* $a \cap b \neq \emptyset$ *(as triples of hexagon points). Then every* $a \in S_A$ *meets exactly* 3 *members of* S_B *and any two members of* S_A *are "joined" by exactly one member of* S_B *— i.e.* Φ *is a Fano plane. The* 21 *points of* P *are then exactly the* flags *of* Φ *(each* $p \in P$ *lies on one line of* S_A *and one of* S_B*, and that pair meets at* p*), and the* 42*-orbit plays the role of the antiflags. This was verified by hand on the explicit spreads (GAP point labels)*

$$S_A = \{\,\{1,12,58\},\{33,37,38\},\{14,26,42\},\{22,34,45\},\{18,29,46\},\{8,11,32\},\{21,40,50\}\,\},$$
$$S_B = \{\,\{1,26,32\},\{8,18,33\},\{12,37,45\},\{14,21,38\},\{34,42,46\},\{29,40,58\},\{11,22,50\}\,\},$$

with all $\binom{7}{2} = 21$ *pairs of* S_A*-members checked to share exactly one concurrent* S_B*-member.*

So: **each of the** 72 **cosets of** K **carries an *ordered* pair of spreads** (S_A, S_B) **whose concurrence geometry is a Fano plane.** We call this a *chiral Fano structure*: the underlying unordered object $\{S_A, S_B\}$ exists in 36 copies, each with two orientations.

6.3 Identification with known objects

The literature search (June 2026) shows the *unordered* structure is classical; the chirality doubling and the Hecke algebra appear to be new.

Theorem 6.4 (The chiral Fano structure is a sub-hexagon of order $(2,1)$)**.** *The incidence structure* $(P,\ S_A \cup S_B)$ *— the* 21*-point orbit together with the* 14 *hexagon lines of the two spreads — is a weak generalized hexagon of order* $(2,1)$*:* 21 *points,* 14 *lines,* 3 *points per line,* 2 *lines per point, incidence graph of girth* 12 *and diameter* 6 *(GAP-verified,* `why13_subhexagon.g`*). Abstractly it is the* dual flag geometry *of the Fano plane: its points are the* 21 *flags of* Φ*, and its* 14 *lines are the* 7 *points and* 7 *lines of* Φ *— the spread* S_A *playing the Fano points,* S_B *the Fano lines. Its incidence graph is the subdivision of the Heawood graph.*

Proof. The parameters and girth/diameter are the GAP computation. For the identification: by Proposition 6.3 each $p \in P$ lies on exactly one line $a(p) \in S_A$ and one $b(p) \in S_B$, and $p \mapsto (a(p), b(p))$ is a bijection $P \to$ flags of Φ (both sets have 21 elements; injectivity because two hexagon lines meet in at most one point). Under this bijection a line $a \in S_A$ becomes the set of flags with first coordinate a — a *pencil*, i.e. a point of Φ together with its three lines — and dually for S_B. This is precisely the dual flag geometry ("double") of the Fano plane, which is the standard thin generalized hexagon of order $(2,1)$: an ordinary hexagon in it alternates point-flags and line-flags around a triangle of Φ, giving girth 12. □

Fact 6.5 (Prior literature)**.**

1. **De Wispelaere–Van Maldeghem (2005)**: the point set P is a *distance*-2 *ovoid* of H(2) (a set of 21 mutually non-opposite... precisely: a maximal coclique-type two-character set meeting every line in 1 or 3 points — compare Proposition 6.2(2)). There are exactly 36 of them, all in one PSU(3,3)-orbit, each with stabilizer $\mathrm{PSL}(2,7) \cong K$; they yield two-weight codes.

2. **De Bruyn–Vandecasteele (2005)**: sub-hexagons of order $(1,q)$ (dually, $(q,1)$) characterize H(q); for $q = 2$ the sub-hexagon is the flag geometry of PG(2,2), as in Theorem 6.4.

3. **Brouwer's tables**: the $U_3(3)$ graph $\mathrm{SRG}(36,14,4,6)$ is the unique rank-3 graph with these parameters, vertices = the 36 copies of $\mathrm{L}_2(7)$ in $\mathrm{U}_3(3)$, point stabilizer $\mathrm{L}_2(7).2$; it is *locally the co-Heawood graph.* Its complement is our $\mathrm{SRG}(36,21,12,12)$ of Proposition 8.2.

4. **Apparently new**: the chirality doubling $36 \to 72$ (the two orientations = the two ways to assign $\{S_A, S_B\}$ to points/lines of Φ, swapped by a *duality* of the Fano plane realized inside $N_G(K) \setminus K$); the rank-8 Hecke algebra $H(G_2(2), \mathrm{PSL}(3,2))$ with its Wedderburn type and the relations $A^2{=}I$, $B^2{=}13I$, $AB{+}BA{=}-4I$; the bipartite graph Γ_3 on $36+36$ with spectrum $\pm 7, (\pm 1)^{21}, (\pm\sqrt{13})^{14}$ and the identity $NN^\top = 7I + 2A_{\mathrm{SRG}}$, i.e. $13 = 7 + 2 \cdot 3$. (Hanaki–Miyamoto's association scheme classification stops far below 72 vertices; the divisible-design-graph tables stop at 39.)

Remark 6.6 (The mirror is a duality)**.** Theorem 6.4 upgrades Proposition 6.7: the normalizing involution t with $S_A^t = S_B$ induces, through the flag bijection, a *correlation* (duality) of the Fano plane Φ. The chirality classes of 36 are therefore "Fano structures with a chosen side of the duality," and $\mathrm{PSU}(3,3)$ vs. its outer coset in $G_2(2)$ is exactly the duality-preserving vs. duality-swapping dichotomy.

Proposition 6.7 (The mirror involution)**.** *The normalizer $N_G(K)$ of order 336 contains an involution $t \notin K$ with $S_A^t = S_B$ (GAP-verified). So the 72 chiral structures fall into 36 mirror pairs $\{(S_A, S_B), (S_B, S_A)\}$, and these mirror pairs are simultaneously the 36 subgroups in the conjugacy class (the two cosets Kg and Ktg of a mirror pair have the same stabilizer $g^{-1}Kg$). The two $\mathrm{PSU}(3,3)$-orbits on cosets from Proposition 6.1 are the two* chirality classes*: $\mathrm{PSU}(3,3)$ preserves orientation; the outer involution of $G_2(2)$ swaps the two halves of 36, pairing the mirrors.*

7 The Meinhold-72 Hecke algebra

7.1 Suborbits and orbitals on the 72 cosets

Proposition 7.1 (GAP-verified)**.** *G acts on the 72 cosets of K with rank 8; suborbit lengths*

$$1,\ 1,\ 7,\ 7,\ 7,\ 7,\ 21,\ 21.$$

In the numbering of `why13_orbitals.g` (which matches the canonical $T_1, \ldots, T_8$ ordering up to the swap $T_7 \leftrightarrow T_8$ noted below): D_1 identity; D_2 the unique nontrivial valency-1 orbital (the mirror pairing *of Proposition 6.7); D_3, D_5 symmetric of valency 7; D_4, D_6 a paired non-symmetric couple of valency 7 ($D_4^\top = D_6$); D_7, D_8 symmetric of valency 21.*

Proposition 7.2 (Geometric meaning of each orbital)**.** *For cosets Kg, Kg', transport the structure (P, S_A, S_B) by the coset representatives and intersect. GAP gives, per orbital:*

orbital	*valency*	$\lvert P \cap P' \rvert$	$\lvert S_A \cap S'_A \rvert$	$\lvert S_A \cap S'_B \rvert$	$\lvert S_B \cap S'_A \rvert$	$\lvert S_B \cap S'_B \rvert$
D_1	1	21	7	0	0	7
D_2	1	21	0	7	7	0
D_3	7	9	1	0	0	3
D_4	7	9	0	1	3	0
D_5	7	9	3	0	0	1
D_6	7	9	0	3	1	0
D_7	21	5	1	0	0	1
D_8	21	5	0	1	1	0

Read off: D_2 is the mirror (same 21 points, spreads swapped). The columns $\lvert S_A \cap S'_A \rvert + \lvert S_B \cap S'_B \rvert > 0$ versus $\lvert S_A \cap S'_B \rvert + \lvert S_B \cap S'_A \rvert > 0$ split $\{D_1, D_3, D_5, D_7\}$ from $\{D_2, D_4, D_6, D_8\}$: ***the first set stays within a chirality class of 36, the second crosses between them.***

Remark 7.3 (Within-half closure)**.** The within-half orbitals $\{D_1, D_4, D_6, D_7\}$? — careful. The chirality split must be checked against the orbital *graphs*, and there it is unambiguous: the graphs of D_3, D_5, D_8 are connected and bipartite (GAP: ± 7 resp. ± 21 in the spectrum), with the two halves of 36 as the parts — these orbitals *cross* between chirality classes. The graphs of D_4, D_6, D_7 have eigenvalue 7 (resp. 21) with multiplicity 2: two connected components, one on each half — these stay *within*. (The invariant table distinguishes ordered structures; graph connectivity is what detects the halves.) Algebraically, the within-half set $\{T_1, T_4, T_6, T_7^{\mathrm{GAP}}\}$ is closed under multiplication: $T_4T_6 = 7T_1 + 2T_8^{\mathrm{can}}$, $T_4T_4 = 4T_6 + T_8^{\mathrm{can}}$, etc.

Remark 7.4 (Numbering dictionary)**.** Canonical (MEINHOLD-HECKE.md) vs GAP suborbit numbering: they agree on $T_1, \ldots, T_6$; the canonical *central* T_8 (with $\rho(T_8) = 3I$) is GAP suborbit 7 (the disconnected 21-valent one), and canonical T_7 is GAP suborbit 8 (the bipartite 21-valent one). From here on we use the **canonical** numbering: T_8 is the 21-valent within-half orbital.

7.2 The algebra and its 2-dimensional irrep

Fact 7.5 (Structure of $H(G,K)$, from MEINHOLD-HECKE.md)**.** $H(G,K) \cong \mathbb{C}^2 \oplus M_2(\mathbb{C}) \oplus \mathbb{C}^2$ with canonical basis $T_1, \ldots, T_8$ as above. Key products (canonical numbering):

$$T_3T_3 = 7T_1 + 2T_8, \quad T_5T_5 = 7T_1 + 2T_8, \quad T_4T_4 = 4T_6 + T_8, \quad T_4T_6 = 7T_1 + 2T_8,$$

$$T_7T_7 = 21T_1 + 12T_4 + 12T_6 + 12T_8.$$

The center is spanned by T_1, $z_3 = T_4 + T_6$, $z_4 = -3T_2 + T_7$, $z_5 = T_8$, and the all-ones combination. In the 2-dimensional irrep ρ:

$$\rho(T_1) = I, \qquad \rho(T_8) = 3I \text{ (central)}, \qquad A := \rho(T_2),\ B := \rho(T_3) = \begin{pmatrix} -2 & 9 \\ 1 & 2 \end{pmatrix},$$

$$A^2 = I, \qquad B^2 = 13I, \qquad AB + BA = -4I.$$

Indeed $B^2 = \rho(T_3T_3) = 7\rho(T_1) + 2\rho(T_8) = 7I + 6I = 13I$.

The question is now sharply posed: **why $T_3^2 = 7T_1 + 2T_8$, and why $\rho(T_8) = 3I$?**

8 Why 13

8.1 Step 1: $T_3^2 = 7T_1 + 2T_8$ by path counting

Proposition 8.1. *In the orbital graph Γ_3 (valency 7, symmetric), for any vertex x: the $\binom{7}{1} \cdot \ldots$ — precisely, the $7 \cdot 7 = 49$ two-step walks from x end: 7 of them back at x, and the remaining 42 are distributed 2 each over the 21 vertices of $D_8(x)$. Equivalently $A_3^2 = 7I + 2A_8$. (GAP path count, suborbits 3 and 5: `[[0,7],1]`, `[[7,2],21]`.)*

Geometrically: x and any Γ_3-neighbour share $|P \cap P'| = 9$ points and one common S_A-line; two structures in relation D_8 ($|P \cap P'| = 5$) can be joined through exactly 2 intermediate structures, and through *no* structure if they are in any other relation. *Why* the values are exactly $(7; 2)$ is the content of the spread combinatorics; we record it as computationally certified and return to a synthetic proof in Open Questions.

8.2 Step 2: Γ_8 is two copies of $\mathrm{SRG}(36, 21, 12, 12)$

Proposition 8.2. *The orbital graph Γ_8 (21-valent, within-half) is the disjoint union of two copies of a strongly regular graph $\mathrm{SRG}(36, 21, 12, 12)$, one on each chirality class of 36. Its spectrum (GAP): $21^2,\ 3^{28},\ (-3)^{42}$; per component $21^1,\ 3^{14},\ (-3)^{21}$.*

Why the eigenvalues are $21, 3, -3$. For an $\mathrm{SRG}(n, k, \lambda, \mu)$ the non-principal eigenvalues are

$$r, s = \frac{(\lambda - \mu) \pm \sqrt{(\lambda - \mu)^2 + 4(k - \mu)}}{2} = \pm\sqrt{k - \mu} \quad \text{when } \lambda = \mu.$$

Here $\lambda = \mu = 12$, $k = 21$, so $r, s = \pm\sqrt{21 - 12} = \pm\sqrt{9} = \pm 3$. The multiplicities $(14, 21)$ follow from trace conditions: $m_r + m_s = 35$, $21 + 3m_r - 3m_s = 0 \Rightarrow m_s - m_r = 7$. □

Remark 8.3. The *complement* of this SRG on each half of 36 is the $U_3(3)$ **graph** $\mathrm{SRG}(36, 14, 4, 6)$ (spectrum $14, 2^{21}, (-4)^{14}$) — a distance-transitive graph on the 36 sub-hexagons/Fano-structure mirror pairs, well known from the ATLAS. This identifies our two halves with a classical object: vertices = the 36 $\mathrm{PSL}(3, 2)$'s of one chirality, adjacency = orbital $D_4 \cup D_6$ (the 7+7-valent directed pair, symmetrized).

8.3 Step 3: assembling 13

Theorem 8.4 (Why 13)**.** $13 = 7 + 2 \cdot 3$*, in the following precise sense. Let N be the 36×36 biadjacency matrix of the bipartite graph Γ_3 between the two chirality classes. Then*

$$NN^\top \ = \ 7I + 2A,$$

where A is the adjacency matrix of $\mathrm{SRG}(36, 21, 12, 12)$ on one half. Consequently

$$\mathrm{spec}(NN^\top) = \{\, 7 + 2 \cdot 21,\ 7 + 2 \cdot 3,\ 7 + 2 \cdot (-3) \,\} = \{49,\ 13^{(14)},\ 1^{(21)}\},$$

and the eigenvalue 13, of multiplicity 14, is the value of B^2 on the isotypic component of the 14-dimensional $G_2(2)$-irreducible χ_{14} — the same irreducible whose multiplicity appears as $m_{-3} = 14$ in the 63-point spectrum (Proposition 3.7). In irrep terms: $B^2 = \rho(T_3)^2 = 7I + 2\rho(T_8) = 7I + 2 \cdot 3I = 13I$, where 3 is precisely the SRG eigenvalue $\sqrt{k - \mu} = \sqrt{21 - 12}$ singled out by the 2-dimensional Wedderburn block.

Proof. Γ_3 is bipartite between the halves (GAP: spectrum contains ± 7; 2-colouring check passes), so its adjacency matrix is $A_3 = \left(\begin{smallmatrix} 0 & N \\ N^\top & 0 \end{smallmatrix}\right)$ and $A_3^2 = \left(\begin{smallmatrix} NN^\top & 0 \\ 0 & N^\top N \end{smallmatrix}\right)$. By Proposition 8.1, $A_3^2 = 7I + 2A_8$, and A_8 is block-diagonal with the SRG on each block (Proposition 8.2); restricting to one block gives $NN^\top = 7I + 2A$. The spectrum follows from Proposition 8.2; in particular the eigenvalues $\pm\sqrt{13}$ of Γ_3, each with multiplicity 14, square to 13 (GAP: characteristic polynomial of Γ_3 is $(x-7)(x+7)(x-1)^{21}(x+1)^{21}(x^2 - 13)^{14}$).

For the irrep statement: T_8 is central, so $\rho(T_8)$ is a scalar θI; the eigenvalues of A_8 are $21, 3, -3$ and the corresponding isotypic dimensions are $2 \cdot 1$ (the $\theta = 21$ piece belongs to the 1-dimensional irreps with $|D_8|$-eigenvalue 21), $28 = 2 \times 14$, and 42. The 2-dimensional irrep occurs in the permutation module $\mathbb{C}[G/K]$ with multiplicity 14 (its multiplicity equals $\deg \chi_{14} = 14$, since $\mathbb{C}[G/K] = \bigoplus_\chi \chi^{\oplus m_\chi}$ with $m_\chi = \langle \chi, \mathrm{ind}_K^G 1 \rangle$ and the Wedderburn block $M_2(\mathbb{C})$ corresponds to $m = 14$), occupying a 28-dimensional isotypic component — exactly the A_8-eigenspace of eigenvalue 3. Hence $\theta = 3$ and $B^2 = 7I + 2 \cdot 3I = 13I$. □

Corollary 8.5 (Restatements of 13). *1.* $13 = 7 + 2\sqrt{k-\mu}$ *for* SRG$(36, 21, 12, 12)$*:* $13 = 7 + 2\sqrt{9}$.

2. Via the complementary $U_3(3)$ *graph* $A_U = J - I - A$*:* $NN^\top = 5I + 2J - 2A_U$*, and on the* χ_{14}*-isotypic component (*$J = 0$*,* A_U*-eigenvalue* -4*):* $13 = 5 + 8 = 5 - 2 \cdot (-4)$*. The eigenvalue* -4 *of the* $U_3(3)$ *graph is the same "*-4*" as in* $AB + BA = -4I$*.*

3. The directed orbitals tell the same story in complex form: Γ_4 *has spectrum* $7^2, 1^{42}, (-2 \pm 3i)^{14}$ *and* $|-2+3i|^2 = 4+9 = 13$*. The non-real eigenvalues are forced by Lemma 5.3 (non-symmetric orbital), and their modulus*^{2} *is again* 13 *because* $T_4T_6 = 7T_1 + 2T_8$ *(i.e.* $\rho(T_4)\overline{\rho(T_4)}^\top$*-type identity at eigenvalue level).*

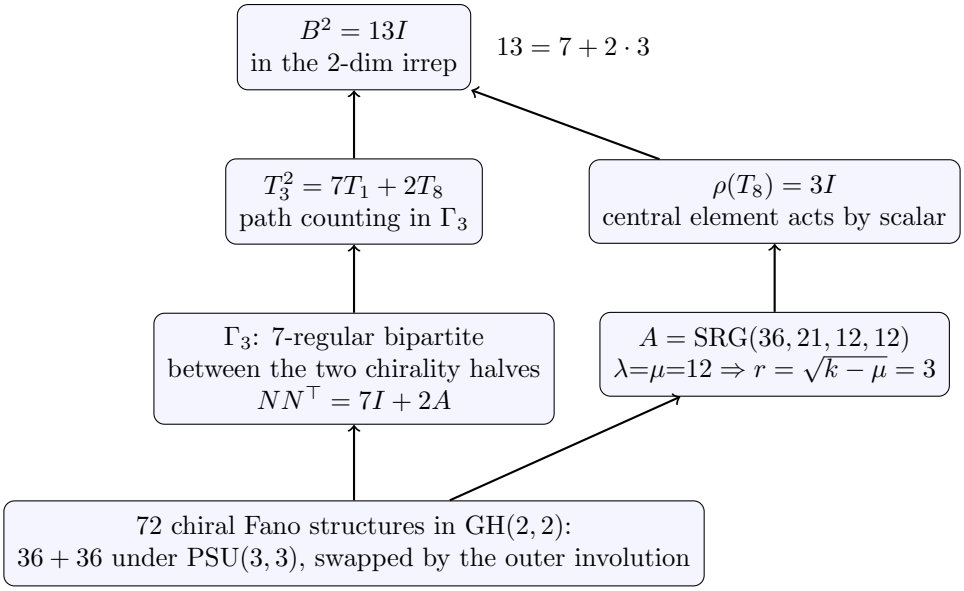


Figure 3: The logical chain behind $B^2 = 13I$.

9 Connections

1. **Fano ↔ hexagon, structurally.** Both are thick generalized polygons of order $(2, 2)$: $n = 3$ and $n = 6$ in the Feit–Higman list. The Fano plane's incidence graph is the Heawood graph (girth 6); the hexagon's incidence graph has girth 12. The hexagon is, in a precise sense, the "girth-doubled" sibling — and by Theorem 6.4 this is not just an analogy: the girth-doubling of the Fano plane (its flag geometry, a thin hexagon of order $(2, 1)$, incidence graph = subdivided Heawood graph) literally embeds in H(2), in exactly 36 ways.

2. **Fano ↔ hexagon, via octonions.** Fano = multiplication chart of compact octonions (Definition 2.11); hexagon = null geometry of split octonions (Definition 4.2). Same exceptional algebra, two forms.

3. **Fano ↔ hexagon, via subgroups.** Aut(Fano) = PSL$(3, 2)$ of order 168 embeds in Aut(H(2)) = $G_2(2)$ of order 12096 with index 72; each embedded copy paints a chiral Fano

structure (two spreads on a 21-point orbit) onto the hexagon, and the 21-orbit is the flag set of that Fano plane: 21 = flags, 42 = antiflags (Fact 2.14 meets Proposition 6.2).

4. **The number 14, three ways.** $\deg \chi_{14} = 14$ is: the multiplicity of eigenvalue -3 in the 63-point collinearity graph; the multiplicity of $\pm\sqrt{13}$ in Γ_3; the multiplicity of the 2-dim irrep in $\mathbb{C}[G/K]$. Also $14 = \dim G_2$ (the Lie algebra!) — χ_{14} is the "adjoint-flavoured" irreducible of $G_2(2)$.

5. **The number 13 elsewhere?** $13 = |\text{points of PG}(2,3)|$ is surely coincidence; but $13 = 14 - 1$ and the χ_{14}-isotypic block carrying eigenvalue 13 has dimension $28 = 2 \cdot 14$ — see Open Questions.

6. **Same theorem, two costumes.** "$B^2 = 13I$" and "the bipartite halved graph of Γ_3 has $NN^\top = 7I + 2A_{\text{SRG}}$" are the *same* statement, once in Wedderburn coordinates, once in adjacency coordinates. Likewise $AB + BA = -4I$ is the $U_3(3)$-graph eigenvalue -4 in Wedderburn coordinates (Corollary 8.5).

10 Open questions

Question 10.1 (Synthetic proof of $T_3^2 = 7T_1 + 2T_8$)**.** Find a purely geometric proof, in the language of spreads in GH(2, 2), that two chiral Fano structures in relation D_8 are connected by exactly 2 structures in relation D_3 to both, while structures in relations D_4, D_6, D_7 are connected by none. A positive answer would replace the GAP certificate in Proposition 8.1 and likely explain the coefficient 2 as "the two points of a hexagon line off a given point" or similar. A negative-style outcome (no clean synthetic argument) would suggest the right level of abstraction is the association scheme, not the geometry.

Question 10.2 (Why $\lambda = \mu$?)**.** The clean form $13 = 7 + 2\sqrt{k-\mu}$ relies on SRG$(36, 21, 12, 12)$ having $\lambda = \mu = 12$, i.e. being a *conference-like* graph in the (λ=μ) family ($k = \lambda + \sqrt{k-\mu}$?). Is there a geometric reason the 36 Fano structures of one chirality form an SRG with $\lambda = \mu$? (E.g. a 2-design structure on the 36.) Partial progress: $\lambda = \mu$ holds iff the SRG eigenvalues are $\pm\sqrt{k-\mu}$, iff the two-dimensional Hecke block has $\operatorname{tr}\rho(T_8') = 0$ for the relevant element — an algebra-level reformulation, not yet an explanation.

Question 10.3 ($13 = 14 - 1$?)**.** Is there natural meaning to $13 = \deg \chi_{14} - 1$, e.g. via the action on the 14-dimensional module with a fixed vector removed? A positive answer would connect $B^2 = 13I$ to the Lie-theoretic $14 = \dim G_2$. A counterexample shape: an analogous Hecke algebra (next q?) where the matching numerology fails.

Question 10.4 (q-analogue)**.** For $G_2(q) \supset \text{PSL}(3, q)$ (when the embedding exists, $q = p \geq 2$): does the double-coset algebra have rank 8, and is there a polynomial $13(q)$ with $13(2) = 13$? Guess shapes: $q^3 + q + \ldots$, or $7 + 2 \cdot 3 \to (q^2{+}q{+}1) + 2{\cdot}$? Computing $q = 3$ ($G_2(3)$, index of PSL$(3, 3) = ?$) is the obvious experiment; it may be too large for naive GAP double-coset enumeration but the orbital method of `why13_orbitals.g` scales.

Question 10.5 ($|P \cap P'| = 9$ coincidence?)**.** For Γ_3-adjacent structures, $|P \cap P'| = 9 = k - \mu$. Same number as in $\sqrt{k-\mu} = 3$. Coincidence or mechanism?

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